

5. To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 100

int main()
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("===== Mini Shell =====\n");
    printf("Type 'exit' to quit.\n\n");

    while (1)
    {
        printf("MiniShell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
            continue;

        if (strcmp(input, "exit") == 0)
            break;

        /* Replace escaped spaces (\ ) with a temporary character */
        for (int i = 0; input[i] != '\0'; i++)
        {
            if (input[i] == '\\' && input[i + 1] == ' ')
            {
                input[i] = 1;          // Temporary marker
                memmove(&input[i + 1], &input[i + 2], strlen(&input[i + 2]) + 1);
            }
        }
    }
}
```

```
    }  
}  
  
int argc = 0;  
char *token = strtok(input, " ");  
  
while (token != NULL && argc < MAX_ARGS - 1)  
{  
    for (int i = 0; token[i] != '\0'; i++)  
    {  
        if (token[i] == 1)  
            token[i] = ' ';  
    }  
  
    args[argc++] = token;  
    token = strtok(NULL, " ");  
}  
  
args[argc] = NULL;  
  
pid_t pid = fork();  
  
if (pid < 0)  
{  
    perror("Fork Failed");  
}  
else if (pid == 0)  
{  
    execvp(args[0], args);  
    perror("Execution Failed");  
    exit(EXIT_FAILURE);  
}  
else  
{  
    wait(NULL);  
}  
}  
  
printf("Shell Closed.\n");  
  
return 0;  
}
```

skill1.c

MiniShell> ls -l

total 128

-rwxr-xr-x	1	thumu_hemanth_kumar	thumu_hemanth_kumar	16168	Aug 14 08:52	skill1
-rw-r--r--	1	thumu_hemanth_kumar	thumu_hemanth_kumar	583	Aug 14 09:28	skill1.c
-rwxr-xr-x	1	thumu_hemanth_kumar	thumu_hemanth_kumar	16000	Aug 14 08:54	skill1b
-rw-r--r--	1	thumu_hemanth_kumar	thumu_hemanth_kumar	188	Aug 14 08:53	skill1b.c
-rwxr-xr-x	1	thumu_hemanth_kumar	thumu_hemanth_kumar	16264	Aug 14 08:57	skill2
-rw-r--r--	1	thumu_hemanth_kumar	thumu_hemanth_kumar	1605	Aug 14 08:56	skill2.c
-rwxr-xr-x	1	thumu_hemanth_kumar	thumu_hemanth_kumar	16264	Aug 29 08:18	skill3
-rw-r--r--	1	thumu_hemanth_kumar	thumu_hemanth_kumar	1668	Aug 29 08:17	skill3.c
-rwxr-xr-x	1	thumu_hemanth_kumar	thumu_hemanth_kumar	16392	Sep 12 08:12	skill4
-rw-r--r--	1	thumu_hemanth_kumar	thumu_hemanth_kumar	2748	Sep 12 08:12	skill4.c
-rwxr-xr-x	1	thumu_hemanth_kumar	thumu_hemanth_kumar	16560	Sep 21 07:01	skill5
-rw-r--r--	1	thumu_hemanth_kumar	thumu_hemanth_kumar	1715	Sep 21 07:01	skill5.c

MiniShell> |